

Tutorial-4, Probability and Statistics (MCC505)
Monsoon Semester: 22-23

1. If 18% of the bolts produced by a machine are defective, determine the probability that out of the 4 bolts chosen at random
 - (i) exactly one will be defective
 - (ii) none will be defective
 - (iii) atmost 2 will be defective
2. In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error.
3. An insurance company has discovered that only about 0.1 per cent of the population is involved in a certain type of accident each year. If its 10,000 policy holders were randomly selected from the population, what is the probability that not more than 5 of its clients are involved in such an accident next year? (hint: use Binomial to Poisson approximation)
4. If the probability that an individual suffers a bad reaction from injection of a given serum is 0.001, determines the probability that out of 2,000 individuals (a) exactly 3, and (b) more than 2 individuals will suffer a bad reaction.
5. One fifth percent of the the blades produced by a blade manufacturing factory turn out to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing no defective, one defective and two defective blades respectively in a consignment of 1,00,000 packets.
6. It is given that 5% of the electric bulbs manufactured by a company are defective. Using poisson distribution find the probability that a sample of 120 bulbs will contain no defective bulb.
7. The probability of a defective steel rod is 0.01. Steel rods are selected at random. Find the probability that the first defect occurs on the ninth steel rod.
8. The literacy rate in a nation is 12%. What is the probability that you ask five persons before one says he/she is literate?

9. An oil company conducts a geological study that indicates that an exploratory oil well should have a 20% chance of striking oil. What is the probability that the first strike comes on the third well drilled?
10. A batch of 10 rocker cover gaskets contains 4 defective gaskets. If we draw samples of size 3 without replacement, from the batch of 10, find the probability that a sample contains 2 defective gaskets.
11. In the manufacture of car tyres, a particular production process is known to yield 10 tyres with defective walls in every batch of 100 tyres produced. From a production batch of 100 tyres, a sample of 4 is selected for testing to destruction. Find: (a) the probability that the sample contains 1 defective tyre (b) the expectation of the number of defectives in samples of size 4 (c) the variance of the number of defectives in samples of size 4.